

CSCE 775: Deep Reinforcement Learning and Search

Domain-Adaptive Synthetic Marine Radar Generation via Reinforcement Learning

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February 13, 2026

1 Introduction

Autonomous surface vessels (ASVs) operating in inland and coastal waterways rely on marine radar for reliable obstacle detection and navigation. Training perception models for these systems demands large quantities of labeled radar data, yet collecting and annotating real-world radar scans across diverse water bodies is expensive, dangerous, and subject to seasonal and geographic constraints. Synthetic radar generators can fill this data gap, but a persistent *sim-to-real domain gap* degrades model performance when transferring from simulated to real environments [Bousmalis et al., 2018, Tobin et al., 2017]. Existing approaches to closing this gap typically rely on hand-tuned generator parameters or static domain randomization, neither of which adapts to the specific characteristics of distinct operating environments such as lakes, rivers, and coastal waters.

We propose an RL-based framework that learns a policy to *automatically* control the parameters of a synthetic marine radar generator so that downstream perception models trained on its output achieve maximal performance on real radar logs. Unlike fixed randomization strategies [Tobin et al., 2017] or one-shot learned distributions [Kar et al., 2019], our approach continuously adapts generator settings conditioned on *water-body domain type* (lake, river, or coast) and live validation feedback. This formulation treats simulator calibration as a sequential decision-making problem, naturally suited to deep reinforcement learning. The result is a closed-loop system that discovers domain-specific synthetic data recipes without manual parameter sweeps, advancing the state of the art in sim-to-real transfer for maritime perception.

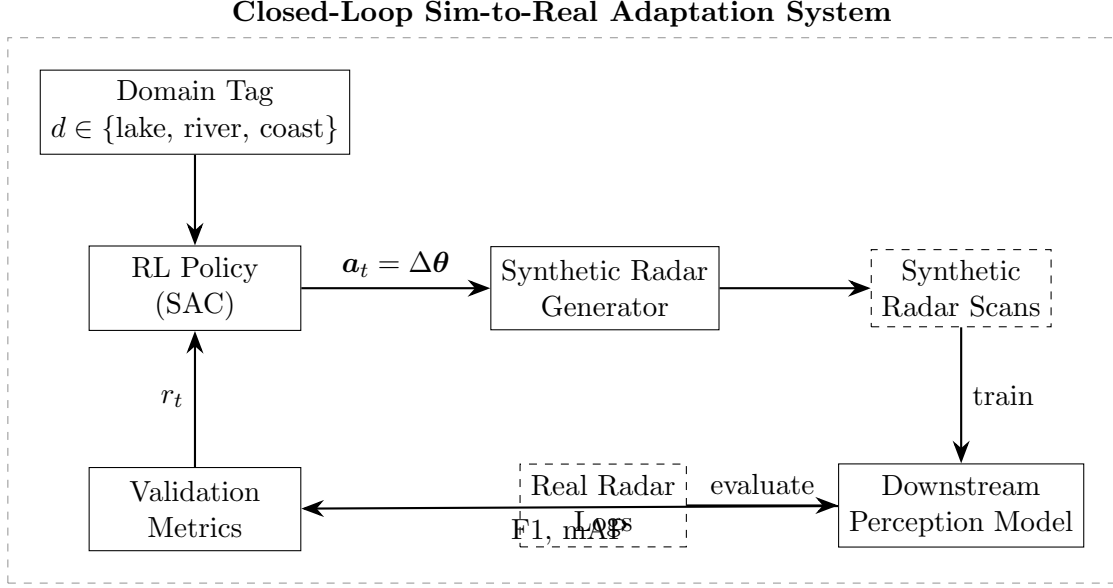


Figure 1: System architecture. The RL policy outputs parameter updates $\Delta\theta$ for the synthetic radar generator, conditioned on domain type. The downstream perception model is trained on synthetic data and evaluated on held-out real radar logs. Validation metrics form the reward signal, closing the loop.

2 Approach

We formulate simulator parameter tuning as a Markov Decision Process (MDP). At each step t , the RL agent observes a state $\mathbf{s}_t$ comprising the current generator parameter vector $\theta_t \in \mathbb{R}^k$, a domain indicator $d \in \{\text{lake, river, coast}\}$, and recent downstream validation metrics. The agent selects a continuous action $\mathbf{a}_t = \Delta\theta_t$ that updates the generator parameters to $\theta_{t+1} = \theta_t + \mathbf{a}_t$. The generator then produces a batch of synthetic radar scans under the new parameters, a downstream perception model is trained (or fine-tuned) on this batch, and evaluated against held-out real radar logs. The reward r_t is the improvement in downstream task score (detection F1 or segmentation mAP) minus a realism penalty that discourages implausible parameter settings.

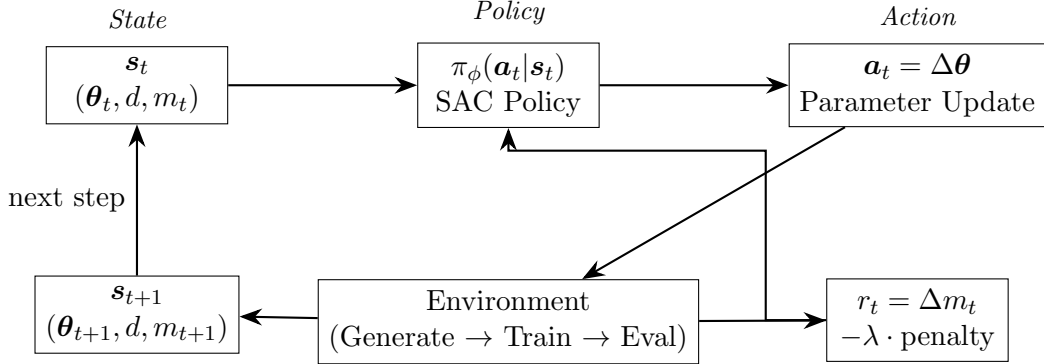


Figure 2: MDP formulation. The state $\mathbf{s}_t$ encodes generator parameters $\boldsymbol{\theta}_t$, domain tag d , and recent metrics m_t . The SAC policy outputs continuous parameter deltas. The environment executes the full generate-train-evaluate loop and returns the next state and reward.

We adopt Soft Actor-Critic (SAC) [Haarnoja et al., 2018] as our primary RL algorithm due to its sample efficiency with off-policy replay and its robust exploration via maximum entropy regularization. The continuous action space is well-suited to SAC, and the entropy bonus naturally encourages diverse generator configurations during early training. We maintain TD3 [Fujimoto et al., 2018] as a secondary algorithm for ablation comparison. The generator parameter vector $\boldsymbol{\theta}$ controls physically meaningful knobs including noise floor, sea clutter intensity, multipath reflection strength, signal attenuation profile, and artifact injection probability. Actions are clipped to physically plausible bounds to maintain radar realism.

Our approach differs from prior work in three key respects. First, unlike static domain randomization [Tobin et al., 2017, Tremblay et al., 2018], our policy adapts generator parameters *online* in response to downstream performance, avoiding wasted computation on uninformative synthetic distributions. Second, unlike Meta-Sim [Kar et al., 2019], which learns a single scene-generation distribution, our agent conditions on water-body type, producing domain-specific synthetic recipes. Third, compared to Active Domain Randomization [Mehta et al., 2020], which searches for maximally informative randomization ranges, our formulation directly optimizes the downstream task objective rather than a proxy transferability metric. The domain-adversarial training framework of Ganin et al. [2016] serves as an additional baseline representing a non-RL approach to domain adaptation.

3 Evaluation

We evaluate on real marine radar logs collected across lake, river, and coastal environments. Data is split by geographic location within each domain to prevent spatial leakage, with 70%/15%/15% train/validation/test splits. The primary downstream task is radar target detection, measured by detection F1 score and mean Average Precision (mAP). We additionally report calibration error and the *sim-to-real gap*, defined as the absolute difference in F1 between a model evaluated on synthetic versus real data.

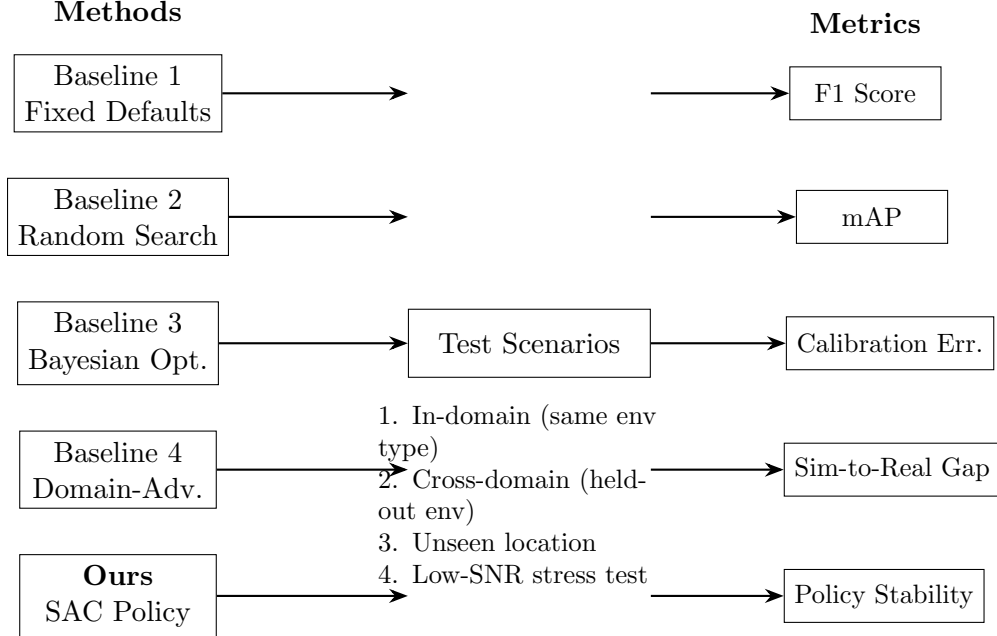


Figure 3: Evaluation pipeline. All methods (four baselines plus our SAC-based approach) are evaluated across four test scenarios. Performance is measured by five complementary metrics capturing both task accuracy and transfer quality.

We compare against four baselines. The first uses the synthetic generator with fixed default parameters. The second replaces the RL policy with random parameter search over the same action space. The third substitutes Bayesian optimization (Gaussian process with expected improvement acquisition) as a non-RL adaptive baseline. The fourth applies domain-adversarial training [Ganin et al., 2016] directly to the perception model without modifying the generator. Evaluation follows four test scenarios of increasing difficulty. *In-domain transfer* trains and tests within the same water-body type. *Cross-domain transfer* trains on two environment types and tests on the held-out third. *Unseen-location robustness* tests on new geographic sites within a known environment type. *Low-SNR stress testing* evaluates under elevated clutter and noise conditions. We additionally conduct ablation studies removing the domain conditioning and the RL loop entirely to isolate the contribution of each component.

4 Conclusion

We propose a reinforcement learning framework for automatically tuning synthetic marine radar generator parameters to minimize the sim-to-real domain gap for downstream perception tasks. By formulating simulator calibration as an MDP and solving it with SAC, our approach replaces manual parameter sweeps with a closed-loop policy that adapts to distinct water-body domains. The evaluation plan spans in-domain, cross-domain, unseen-location, and stress-test scenarios, compared against fixed-parameter, random search, Bayesian optimization, and domain-adversarial baselines across detection F1, mAP, calibration error, sim-to-real gap, and policy stability metrics. If successful, this work will demonstrate that RL-driven simulator control offers a principled, automated path to high-fidelity synthetic

training data for maritime autonomy, reducing the need for costly real-world data collection campaigns.

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